

# Learning Safe Policies from Offline Data under Dynamic Safety Budgets

A Constraint-Conditioned Approach to Offline Safe RL

RL Mid-term Presentation

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# Background: Why Offline RL?

Traditional reinforcement learning usually requires repeated interaction between an agent and the environment.

In many real-world domains, online exploration is expensive, risky, or unacceptable.

## **Robotics**

Avoid collisions, falls, and unsafe motions

## **Autonomous Driving**

Unsafe exploration cannot be used to collect data

## **Healthcare**

Policies must satisfy strict risk budgets

# From Offline RL to Offline Safe RL

Offline RL aims to learn a policy only from a fixed dataset, without additional environment interaction.

In safety-critical settings, the policy must also satisfy a cumulative cost constraint:

$$\max_{\pi} \mathbb{E}_{\pi} \left[ \sum_t r_t \right] \quad \text{s.t.} \quad \mathbb{E}_{\pi} \left[ \sum_t c_t \right] \leq \epsilon$$

where:

- $r_t$ : task reward, such as speed, distance, or completion
- $c_t$ : safety cost, such as collision, boundary violation, or overspeeding
- $\epsilon$ : a fixed safety threshold for the total acceptable risk

# New Challenges in Offline Safe RL

## 1. Offline distribution shift

The dataset only covers historical behavior. At deployment, the learned policy may visit rare or unsupported state-action pairs, making value estimation unreliable.

## 2. Fixed safety constraints

Many Safe RL methods train for a fixed threshold  $\epsilon$ . In deployment, the available safety budget may vary across scenarios, tasks, or time steps.

The core tension: maximize return while avoiding unsafe extrapolation and budget violation.

# Our Research Question

Can we learn a policy from offline data that, at deployment time, can:

- Achieve high reward
- Satisfy safety constraints
- Adapt to different safety budgets
- Reduce the risk from OOD states and actions

$$\pi_{\theta}(a_t \mid s_t, \kappa_t)$$

**In short: the policy should adapt its behavior according to the current remaining safety budget.**

# Core Idea: Condition on the Safety Budget

Instead of training a separate policy for each fixed threshold  $\epsilon$ , we condition the policy on the current remaining budget:

$$\pi_{\theta}(a_t \mid s_t, \kappa_t)$$

Here,  $\kappa_t$  denotes the remaining safety budget at time step  $t$ . Given an initial budget  $\kappa_1$ , it is updated after each step:

$$\kappa_{t+1} = \kappa_t - c_t$$

$$\underbrace{s_t}_{\text{state}}, \quad \underbrace{\kappa_t}_{\text{remaining safety budget}} \quad \longrightarrow \quad \underbrace{\pi_{\theta}(a_t \mid s_t, \kappa_t)}_{\text{constraint-conditioned actor}} \quad \longrightarrow \quad \underbrace{a_t}_{\text{action}}$$

$$a_t \quad \Longrightarrow \quad \begin{cases} R = \sum_{t=0}^T r_t, & \text{maximize reward return} \\ C = \sum_{t=0}^T c_t \leq \epsilon, & \text{satisfy total cost threshold} \end{cases}$$

Changing the initial budget  $\kappa_1$  allows the same policy to trade off reward and cost adaptively.

# Method: Constraint-Conditioned Actor-Critic

The training pipeline can be summarized in five modules:

| Module                      | Role                                            | Notation                                              |
|-----------------------------|-------------------------------------------------|-------------------------------------------------------|
| Offline data                | Build budget-annotated transitions              | $D = \{(s_t, a_t, r_t, c_t, s_{t+1}, \kappa_t)\}$     |
| Constraint-conditioned CVAE | Generate state-action pairs under a budget      | $(\hat{s}, \hat{a}) \sim p_\phi(s, a \mid z, \kappa)$ |
| OOD classifier              | Detect whether $(s, a, \kappa)$ is OOD / unsafe | $h_\psi(s, a \mid \kappa) \in [0, 1]$                 |
| Cost critic                 | Overestimate cost values for OOD samples        | $Q_c(s, a \mid \kappa)$                               |
| Reward critic + Actor       | Maximize reward under safety constraints        | $Q_r(s, a \mid \kappa), \pi_\theta(a \mid s, \kappa)$ |

$$D \rightarrow p_\phi(s, a \mid z, \kappa) \rightarrow h_\psi(s, a \mid \kappa) \rightarrow Q_c(s, a \mid \kappa) \rightarrow \pi_\theta(a \mid s, \kappa)$$

Take-away: OOD / unsafe samples are assigned high cost, so the actor learns to avoid high-risk actions.

# Feasibility and Experimental Plan

We plan to evaluate the framework on the DSRL benchmark.

| Dimension    | Plan                                                                         |
|--------------|------------------------------------------------------------------------------|
| Environments | Run / Circle / Velocity-style safety control tasks                           |
| Metrics      | Normalized reward, normalized cost, constraint violation                     |
| Baselines    | BC-safe, CQL-based Safe RL, Lagrangian Offline RL, sequence modeling methods |
| Ablations    | Remove CVAE, remove OOD classifier, fixed vs. varying safety budgets         |

Feasibility: the datasets and environments are available; Actor-Critic, CVAE, and classifier modules are standard; by the final presentation, we can first reproduce the main reward-cost trade-off and then add one extension such as dynamic budget switching or sparse safe trajectories.



# Related Work

| Direction             | Methods                         | What they address                       | Limitation                                |
|-----------------------|---------------------------------|-----------------------------------------|-------------------------------------------|
| Offline RL            | CQL, IQL, BCQ, BEAR             | Distribution shift                      | No explicit safety constraints            |
| Safe RL               | Lagrangian, CPO, PID-Lagrangian | Online constrained optimization         | Needs interaction; often fixed thresholds |
| Offline Safe RL       | CPQ, COptiDICE, VOCE, FISOR     | Reward-cost trade-off from offline data | Hard to adapt to changing budgets         |
| Sequence / trajectory | CDT, TREBI                      | Condition on reward / cost              | Limited generalization to unseen budgets  |

**Key gap: adapting behavior under dynamic safety budgets.**

Q&A

Thank you!

# References

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